

Uniform Hölder Nonexpansive Maps on Decomposable Banach Balls

Automatic Math Discovery Run `fa.banach.001`

June 12, 2026

Source

Cleon S. Barroso and Valdir Ferreira, *Hölder-contractive mappings, nonlinear extension problem and fixed point free results*, arXiv:2207.03057.

The final open questions ask whether B_X fails the fixed point property for uniformly α -Hölder nonexpansive maps with null minimal displacement when, among other cases, X is isomorphic to a Hilbert space or X is reflexive with an unconditional basis.

5.1 Open questions

We end this paper with the following questions: Let X be an infinite-dimensional Banach space. Does B_X fail the FPP for *uniformly* α -Hölder nonexpansive maps with null minimal displacement if, (1) X is reflexive, (2) X is isomorphic to a Hilbert space or (3) X is reflexive and has an unconditional basis, or then (4) X is the James's space J_2 ? It should be noted that, intuitively, the general strategy used here to handle (Q3) was to check the failure of the HFPP in a certain subset K of B_X and then try to properly extend it to the whole ball. In some cases this involves the availability of extension results with prescribed conditions, which seems to be the main difficulty in dealing with such a problem.

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Result

Theorem 1. *Let X be an infinite-dimensional Banach space that splits as*

$$X = Y \oplus Z$$

with Y and Z closed infinite-dimensional subspaces. Then, for every $\alpha \in (0, 1)$, there is a map $V : B_X \rightarrow B_X$ such that

$$\text{Fix}(V) = \emptyset, \quad d(V, B_X) = 0,$$

and

$$\|V^n x - V^n y\| \leq \|x - y\|^\alpha \quad (x, y \in B_X, \ n \geq 1).$$

Thus V is uniformly α -Hölder nonexpansive.

Corollary 1. *For every $\alpha \in (0, 1)$, the source paper's final open questions have affirmative answers in the following cases:*

1. *X is isomorphic to an infinite-dimensional Hilbert space;*
2. *X has an unconditional basis, in particular when X is reflexive and has an unconditional basis.*

Input Lemma From The Run

We use the decomposable-space packet already established in this run: if $X = Y \oplus Z$ with both summands infinite-dimensional, then there is a map $U : B_X \rightarrow B_X$ such that

$$\text{Fix}(U) = \emptyset, \quad d(U, B_X) = 0, \quad M := \sup_{n \geq 1} \text{Lip}(U^n) < \infty.$$

This is the result recorded in

`solutions/partial/2307.12958.decomposable.tube.uniformly.lipschitz.q2/.`

The present packet only adds the Hölder-nonexpansive scaling step.

Scaling Lemma

Lemma 1. *Let X be a Banach space. Suppose $U : B_X \rightarrow B_X$ satisfies*

$$\text{Fix}(U) = \emptyset, \quad d(U, B_X) = 0, \quad M := \sup_{n \geq 1} \text{Lip}(U^n) < \infty.$$

Then for every $\alpha \in (0, 1)$ there is a uniformly α -Hölder nonexpansive map $V : B_X \rightarrow B_X$ with $\text{Fix}(V) = \emptyset$ and $d(V, B_X) = 0$.

Proof. Let $C = 2M$ and put

$$\eta = \min\{2, C^{-1/(1-\alpha)}\}.$$

Choose $r > 0$ so small that $2r \leq \eta^\alpha$. In particular $r < 1$. Let $R_r : B_X \rightarrow rB_X$ be the radial retraction

$$R_r(x) = \begin{cases} x, & \|x\| \leq r, \\ r x / \|x\|, & \|x\| > r. \end{cases}$$

This retraction is 2-Lipschitz on every normed space.

Define the scaled map $S : rB_X \rightarrow rB_X$ by

$$S(z) = r U(z/r).$$

Then $S^n(z) = r U^n(z/r)$, so $\text{Lip}(S^n) \leq M$ for every $n \geq 1$. Moreover, S is fixed-point free and has null minimal displacement on rB_X .

Now set

$$V = S R_r : B_X \rightarrow B_X.$$

Since $S(rB_X) \subseteq rB_X$ and R_r is the identity on rB_X ,

$$V^n = S^n R_r \quad (n \geq 1).$$

For $x, y \in B_X$, write $d = \|x - y\|$. Then

$$\|V^n x - V^n y\| \leq \min\{M\|R_r x - R_r y\|, 2r\} \leq \min\{Cd, 2r\}.$$

If $d \leq \eta$, then $Cd \leq d^\alpha$ by the definition of η . If $d \geq \eta$, then $2r \leq \eta^\alpha \leq d^\alpha$. Hence

$$\|V^n x - V^n y\| \leq d^\alpha$$

for all $x, y \in B_X$ and every $n \geq 1$.

There is no fixed point: if $Vx = x$, then $x \in rB_X$, so $R_r x = x$ and $Sx = x$, contradicting $\text{Fix}(S) = \emptyset$.

Finally, since S has null minimal displacement on rB_X , there are $z_j \in rB_X$ such that $\|Sz_j - z_j\| \rightarrow 0$. For these points $R_r z_j = z_j$, so $\|Vz_j - z_j\| \rightarrow 0$. Thus $d(V, B_X) = 0$. $\square$

Proof of Theorem 1. Apply the input lemma from the run to obtain U , then apply Lemma 1. $\square$

Proof of Corollary 1. If X is isomorphic to an infinite-dimensional Hilbert space, split the Hilbert space into two closed infinite-dimensional orthogonal summands and transport that decomposition to X through an isomorphism. The transported summands are closed and complemented, so Theorem 1 applies.

If X has an unconditional basis, split the basis into its odd and even subsequences. The associated coordinate subspaces are closed, infinite-dimensional, and complemented by the unconditional coordinate projections. Again Theorem 1 applies. The reflexive unconditional-basis case from the source paper is therefore included. $\square$

Limitations

This does not answer the source paper's arbitrary reflexive case, nor the James-space case. The argument relies on the existence of a decomposable two-infinite-summand structure, because that is what supplies the fixed-point-free uniformly Lipschitzian dynamics with null minimal displacement.

Verification Notes

The only scalar estimate in the new argument is

$$\min\{Cd, 2r\} \leq d^\alpha \quad (0 \leq d \leq 2),$$

with $\eta = \min\{2, C^{-1/(1-\alpha)}\}$ and $2r \leq \eta^\alpha$. The accompanying script `code/verify_scaling_lemma.py` samples this inequality for representative values of C , r , and α .

Human Review: Uniform Hölder nonexpansive maps on decomposable balls

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct and the method is sound. The key point is that a previous packet in the run had already produced the required uniformly Lipschitzian fixed-point-free map with null minimal displacement on B_X for every decomposable Banach space

$$X = Y \oplus Z,$$

where both summands are infinite-dimensional. Once this input is available, the present argument is a clean scaling step which converts uniformly Lipschitzian dynamics into uniformly α -Hölder nonexpansive dynamics.

There is also a bibliographic typo in the packet: the source paper is listed as being by Cleon S. Barroso and Valdir Ferreira, whereas the paper *Hölder-contractive mappings, nonlinear extension problem and fixed point free results* is authored only by Cleon S. Barroso.

Scaling Argument

Starting with $U : B_X \rightarrow B_X$ such that

$$\text{Fix}(U) = \emptyset, \quad d(U, B_X) = 0, \quad M := \sup_{n \geq 1} \text{Lip}(U^n) < \infty,$$

the proof compresses the dynamics into a small ball rB_X . The scaled map is

$$S(z) = rU(z/r), \quad z \in rB_X.$$

It has the same fixed-point-free and null-minimal-displacement properties, and its iterates remain uniformly Lipschitzian.

The map on the whole unit ball is then

$$V = SR_r,$$

where $R_r : B_X \rightarrow rB_X$ is the radial retraction. Since R_r is 2-Lipschitz and $S(rB_X) \subseteq rB_X$, one obtains

$$V^n = S^n R_r$$

and therefore

$$\|V^n x - V^n y\| \leq \min\{2M\|x - y\|, 2r\}.$$

Choosing r sufficiently small gives

$$\min\{2Md, 2r\} \leq d^\alpha \quad (0 \leq d \leq 2),$$

which is exactly the uniformly α -Hölder nonexpansive estimate

$$\|V^n x - V^n y\| \leq \|x - y\|^\alpha \quad (n \geq 1).$$

Scope of the Result

The automatic packet correctly observes that the result applies to spaces isomorphic to Hilbert space and to spaces with an unconditional basis, because these are decomposable in the required sense.

However, the packet missed one important consequence: the James space case is also covered. The James space is decomposable, and therefore it falls under the same theorem. Thus the argument should be recorded not only as answering the Hilbert-isomorphic and reflexive-unconditional-basis cases, but also as giving the corresponding affirmative answer for James space.

Review Outcome

Archive this packet as an apparently correct solution for all decomposable Banach spaces covered by the input uniformly Lipschitzian construction. In particular, it gives affirmative answers in the Hilbert-isomorphic case, the unconditional-basis case, and the James-space case. The remaining genuinely unresolved part, from the perspective of this packet, is the arbitrary reflexive case without an available decomposable uniformly Lipschitzian construction.